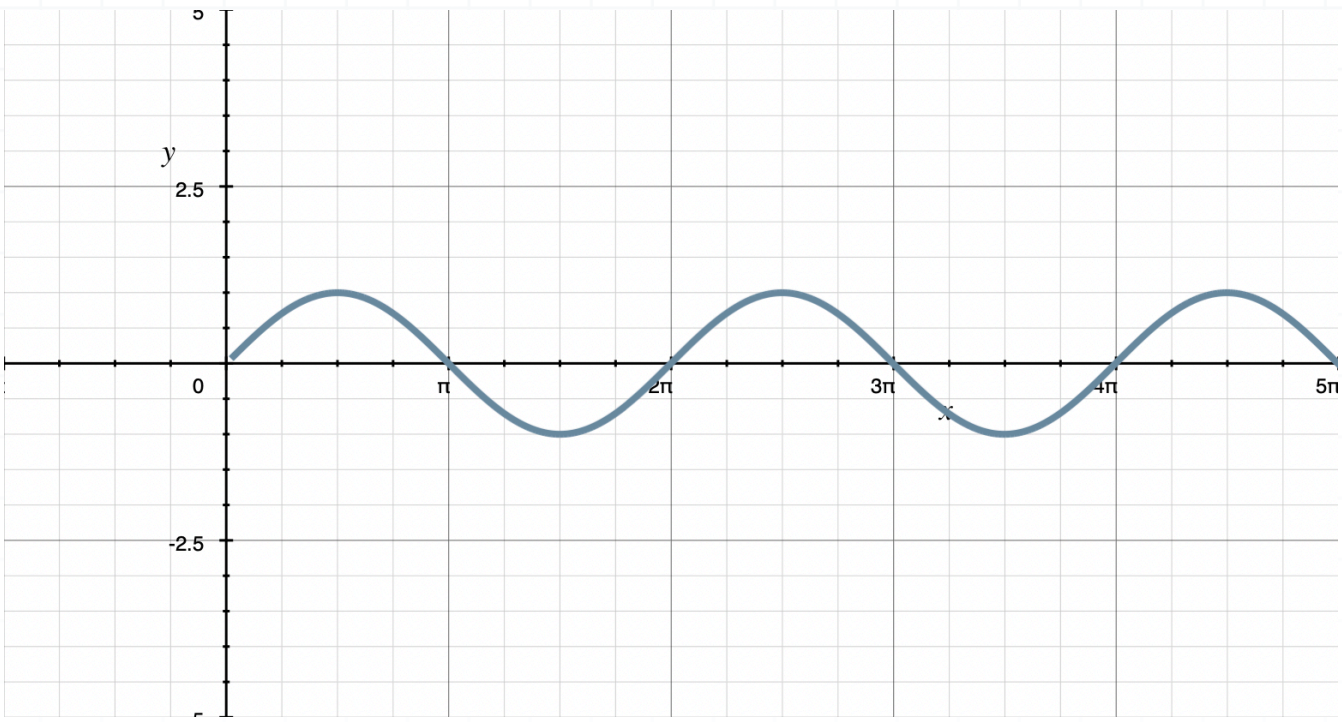
Second Shifting Theorem

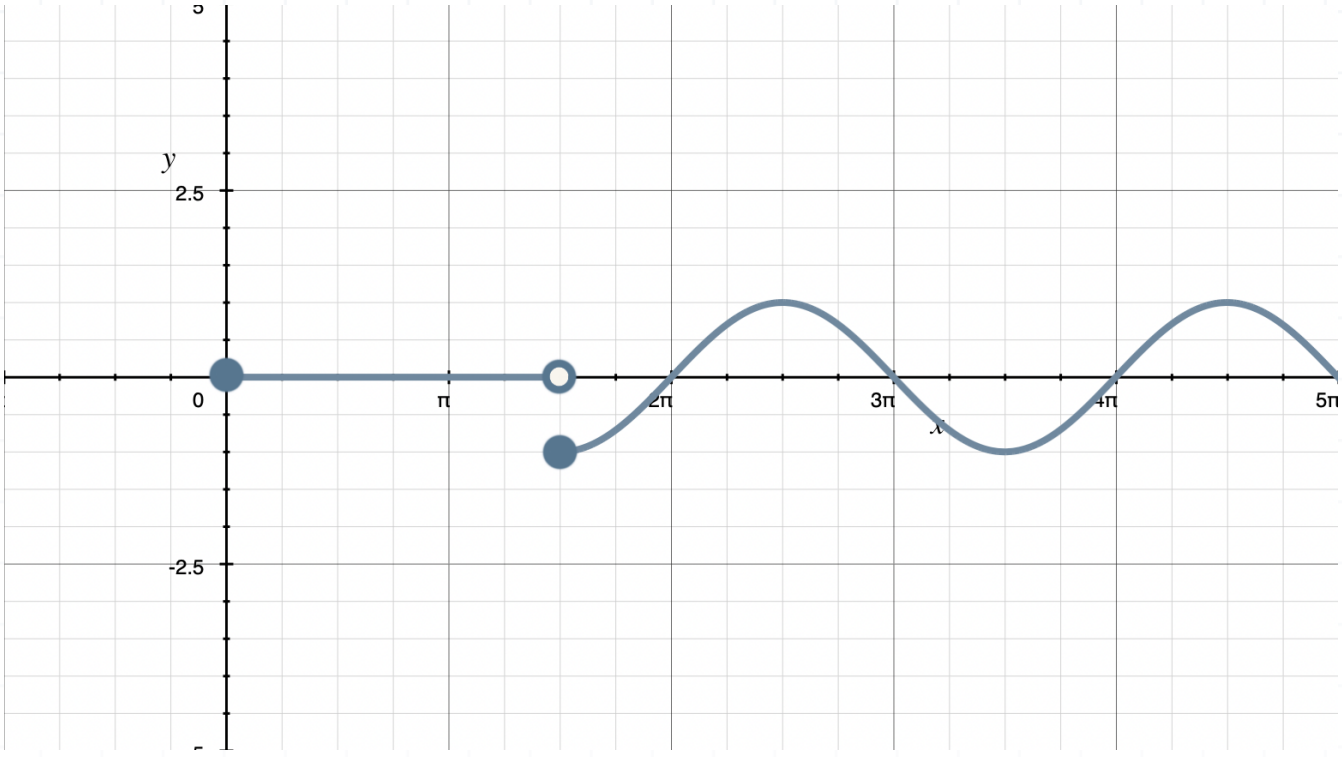
Below is the graph of the sine function $f(t) = \sin t$ for $t \geq 0$.



Previously we looked at how to turn off the part of this sine function on $0 \leq t < 3\pi/2$, by multiplying by a unit step function.

$$u(t-c)f(t) = u\left(t - \frac{3\pi}{2}\right)\sin t$$

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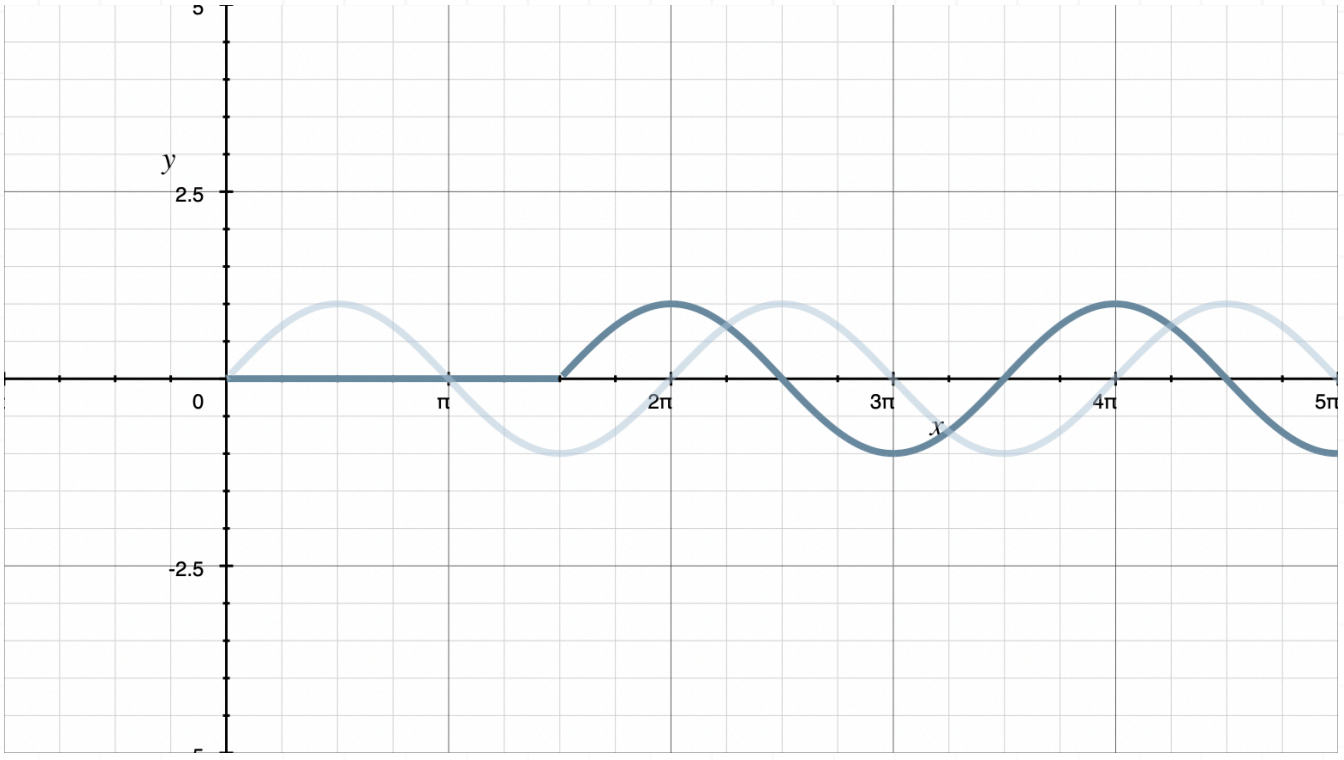


We said that, in general, we can turn off any function $f(t)$ on the interval $0 \leq t < c$ by multiplying $f(t)$ by $u(t-c)$ or $u_c(t)$.

But now we want to look at a slightly different way to turn off $f(t)$. This time, instead of replacing $f(t) = \sin t$ with $f(t) = 0$ on the interval $0 \leq t < 3\pi/2$, we want to shift the part of $f(t) = \sin t$ for $t \geq 0$ to the right a distance of $t = c$.

In other words, we still want the graph "off" on $0 \leq t < 3\pi/2$, but once we get to $t = c = 3\pi/2$ we want to see all of $f(t) = \sin t$, as if we were starting $f(t) = \sin t$ at $t = 0$, like this:

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In this graph, the start of $f(t) = \sin t$ just slid over to the right from $t = 0$ to $t = 3\pi/2$.

This kind of shift is important to understand because it'll come up frequently when we start applying the Laplace transform to step functions.

Second Shifting Theorem

To represent a function that's shifted over by some distance after being "turned off" over that same distance, we use

$$f(t-c)u(t-c) = \begin{cases} 0 & 0 \le t < c \\ f(t-c) & t \ge c \end{cases}$$

Using the example above, where we shifted the $t \ge 0$ part of the sine function to the right by $3\pi/2$, and turning off the function on $0 \le t < 3\pi/2$, we would express this as

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$$\sin\left(t-\frac{3\pi}{2}\right)u\left(t-\frac{3\pi}{2}\right)=\begin{cases}0 & 0\le t<\frac{3\pi}{2}\\ \sin\left(t-\frac{3\pi}{2}\right) & t\ge\frac{3\pi}{2}\end{cases}$$

The Laplace transform of the left side of this equation, which takes the general form $f(t-c)u(t-c)$, is what's given by the **Second Shifting Theorem**, or Second Translation Theorem:

$$\mathscr{L}(f(t-c)u(t-c))=e^{-cs}F(s)$$

when $F(s)=\mathscr{L}(f(t))$ and $c>0$. By this theorem, the Laplace transform of the sine function we've been working with would be

$$\mathscr{L}\left(\sin\left(t-\frac{3\pi}{2}\right)u\left(t-\frac{3\pi}{2}\right)\right)=e^{-\frac{3\pi}{2}s}\frac{1}{s^2+1}$$

because $1/(s^2+1)$ is the Laplace transform of $\sin t$. Let's do another example where we apply a Laplace transform using the Second Shifting Theorem.

Example

Use a step function to represent shifting the portion of $f(x)$ for $x\ge 0$ to the right 3 units, while turning off the function on $0\le x<3$. Then take its Laplace transform.

$$f(x)=\frac{1}{2}x^3+x^2-x$$

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Because we want to shift the curve 3 units to the right, we can start by substituting $c = 3$ into our formula for this kind of shift.

$$f(x-c)u(x-c) = \begin{cases} 0 & 0 \le x < c \\ f(x-c) & x \ge c \end{cases}$$

$$f(x-3)u(x-3) = \begin{cases} 0 & 0 \le x < 3 \\ f(x-3) & x \ge 3 \end{cases}$$

Now we just need to replace $f(x-3)$, which is

$$f(x) = \frac{1}{2}x^3 + x^2 - x$$

$$f(x-3) = \frac{1}{2}(x-3)^3 + (x-3)^2 - (x-3)$$

So the shifted function is represented by

$$\left(\frac{1}{2}(x-3)^3 + (x-3)^2 - (x-3)\right)u(x-3)$$

$$= \begin{cases} 0 & 0 \le x < 3 \\ \frac{1}{2}(x-3)^3 + (x-3)^2 - (x-3) & x \ge 3 \end{cases}$$

To find the Laplace transform of this function, we'll need the Laplace transform of $f(x)$.

$$F(s) = \mathcal{L}(f(x)) = \mathcal{L}\left(\frac{1}{2}x^3\right) + \mathcal{L}(x^2) - \mathcal{L}(x)$$

$$F(s) = \mathcal{L}(f(x)) = \frac{1}{2}\mathcal{L}(x^3) + \mathcal{L}(x^2) - \mathcal{L}(x)$$

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$$F(s) = \mathscr{L}(f(x)) = \frac{1}{2}\left(\frac{3!}{s^{3+1}}\right) + \frac{2!}{s^{2+1}} - \frac{1!}{s^{1+1}}$$

$$F(s) = \mathscr{L}(f(x)) = \frac{3}{s^4} + \frac{2}{s^3} - \frac{1}{s^2}$$

Then by the Second Shifting Theorem, the Laplace transform of the shifted function is

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-cs}F(s)$$

$$\mathscr{L}(f(t-c)u(t-c)) = e^{-3s}\left(\frac{3}{s^4} + \frac{2}{s^3} - \frac{1}{s^2}\right)$$

Proving the SST

To prove the Second Shifting Theorem, we can start with the definition of the Laplace Transform,

$$\mathscr{L}(u_c(t)f(t-c)) = \int_0^\infty e^{-st}u_c(t)f(t-c)\ dt$$

Remember that for $0 \le t < c$, the value of the step function is $u_c(t) = 0$, which means the value of the integral on the interval $0 \le t < c$ will be 0. So we only really need to consider the integral from the point at which $t = c$, and we can therefore change the lower limit of integration from $t = 0$ to $t = c$. Once we do, we know $u_c(t) = 1$, which means the step function simplifies out of the integral.

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$$\mathscr{L}(u_c(t)f(t-c)) = \int_c^\infty e^{-st} f(t-c)\ dt$$

Make a substitution $v = t - c$ and $dv = dt$. Plugging the limits of integration $t = c$ and $t = \infty$ into $v = t - c$ gives new limits of integration $v = 0$ and $v = \infty$.

$$\mathscr{L}(u_c(t)f(t-c)) = \int_0^\infty e^{-s(v+c)} f(v)\ dv$$

$$\mathscr{L}(u_c(t)f(t-c)) = \int_0^\infty e^{-sv-sc} f(v)\ dv$$

$$\mathscr{L}(u_c(t)f(t-c)) = \int_0^\infty e^{-sv} e^{-sc} f(v)\ dv$$

$$\mathscr{L}(u_c(t)f(t-c)) = e^{-sc} \int_0^\infty e^{-sv} f(v)\ dv$$

The remaining integral is the definition of the Laplace transform of $f(v)$, so

$$\mathscr{L}(u_c(t)f(t-c)) = e^{-sc} F(s)$$

This formula works as long as the function $f(t)$ is shifted by c. In other words, as long as $v = t - c$. This is how we get to the formula for the Second Shifting Theorem.

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